

Sustainable Marine Biofouling Control: From Biocide-Free Nanocomposite and Bioinspired Coatings to Smart Cleaning, Hydrodynamic Performance, and Environmental Risk

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Abstract—Marine biofouling is a coupled surface, biological, hydrodynamic, environmental, and maintenance problem. This critical review synthesizes an 81-paper curated corpus on antifouling and fouling-release coatings, nanomaterial-enabled surfaces, bioinspired and biobased interfaces, active cleaning, ship performance, and environmental risk. The evidence supports a distinction between preventing settlement, reducing adhesion, releasing established fouling, and physically removing fouling; these modes are not interchangeable and their performance depends on organism, exposure, motion, ageing, and cleaning conditions. Nanocomposites and functional interfaces frequently improve laboratory antifouling or adhesion metrics, but dispersion, wear, immersion stability, and constituent release remain deployment constraints. Ship-scale studies show that biofilm and roughness can impose drag and fuel penalties, while cleaning can introduce coating wear and contaminated effluent. The most defensible engineering pathway is therefore a system architecture combining low-adhesion surfaces, controlled active functions where justified, condition-based cleaning, hydrodynamic validation, and life-cycle assessment. The corpus remains heterogeneous, with limited long-duration full-scale and comparable ecotoxicological evidence.

Index Terms—marine biofouling, fouling-release coatings, nanocomposites, slippery surfaces, hull cleaning, hydrodynamic drag, environmental risk, life-cycle assessment

I. INTRODUCTION

A. Marine biofouling as an engineering problem

Marine biofouling begins with conditioning films and microbial attachment and can progress to diatoms, algae, invertebrates, and mixed communities. The consequence is not simply visual contamination: attached material changes roughness, interfacial shear, mass, cleaning demand, and the probability of transporting non-indigenous organisms. The source corpus spans ship hulls, aquaculture nets, submerged coupons, heat-exchanger-like systems, and cleaning equipment; these assets should not be treated as equivalent.

B. Operational, hydrodynamic, and environmental consequences

For ships, biofilm and macrofouling alter the effective hull surface and can increase resistance and powering demand. Operating-ship and modelling studies connect fouling growth,

coating quality, hull roughness, fuel use, and dry-docking decisions [1]–[4]. In-water cleaning can restore surface condition but may abrade coatings and mobilize paint particles or fouling organisms [5]–[7]. Thus a coating that performs well in a static coupon test is not automatically sustainable at ship scale.

C. From toxic biocides to integrated control

The corpus reflects a transition from conventional metal- and biocide-containing systems toward silicone fouling-release coatings, low-adhesion and lubricated interfaces, photocatalytic or UV-assisted functions, bio-derived compounds, and mechanical management. Biocide-free does not mean impact-free: nanomaterial release, coating wear, cleaning effluent, manufacturing burden, and service life remain relevant environmental dimensions.

D. Review question and objectives

This review asks how sustainable marine biofouling control can integrate lower-impact coatings, nanomaterial-enabled and bioinspired surfaces, smart cleaning, and engineering-scale maintenance while addressing durability, hydrodynamics, environmental release, and cost. It evaluates: (i) mechanism-to-fouling relationships; (ii) the durability and validation boundary of functional surfaces; (iii) the operational role of cleaning; and (iv) the evidence needed to move from material screening to asset-level decisions.

E. Positioning against overlapping reviews

This manuscript does not claim novelty from being the first review of sustainable marine antifouling materials. Qiu et al. provide a broad functional-polymer review organized around surface properties and coating functions, while Pourhashem et al. review antifouling nanocomposite polymers through experimental, mechanistic, and theoretical lenses [8], [9]. Carbon-focused work, including graphene-based marine biofilm mitigation, addresses a narrower material and biofilm problem [10]. Recent reviews by Roumeu et al. and Sun et al. have focused on nanotechnology and multifunctional bioinspired surfaces; these reinforce why material taxonomy alone is not a sufficient contribution. The contribution here is different in level and method: the 81-study curated corpus is screened across coatings, surfaces, cleaning, hydrodynamics, release, economics,

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and environmental risk; evidence is explicitly weighted with a reproducible SRI rubric and high-SRI sensitivity analysis; and material findings are translated into ISO 19030/ITTC performance variables, condition-based cleaning decisions, standards-aligned release testing, and regulatory boundaries. The result is a cross-scale engineering synthesis rather than another taxonomy of coating chemistries. These distinctions are scope and evidence-treatment contributions, not a claim that the underlying material mechanisms are unprecedented.

II. REVIEW SCOPE, EVIDENCE BASE, AND SYNTHESIS METHOD

A. Scope and source audit

The evidence base comprises an 81-study curated corpus. Each study was screened by title and available abstract, conclusion, or closing discussion. Fifty-one studies are classified as core because they directly address marine fouling control, coatings, surfaces, hulls, or submerged infrastructure; twenty-six are supporting because they address hydrodynamics, cleaning, environmental effects, economics, or biofilm mechanics; two are peripheral methodological or contextual studies; and two are excluded from the core narrative because their primary systems are not marine fouling control. The complete audit is supplied in Supplementary Material S1; exclusion is retained rather than hidden.

B. Database search, time window, and selection funnel

To systematically establish the evidence boundaries and query precision, a tiered Scopus search was conducted (database: Scopus; time window: 2015–2026; searched 5 September 2026) using eight Boolean query tiers targeting the strict intersection of biomass-derived pyrogenic-carbon nanomaterials with marine antifouling (Tier 1, strict and broadened variants), adjacent synthesis and coatings pillars (Tier 2), adjacent marine-nanomaterial, hull fouling-release, and aquaculture pillars (Tier 3), and green/sustainable antifouling literature (Tier 4). The raw search returned 21,098 paper records and 2,447 review records across the eight tiers, as reported in Supplementary Material S2. After per-tier fetch caps and de-duplication, 1,160 unique records were exported for title-level screening.

Title-level screening of the strict Tier-1 intersection (200 records screened) retained only 1 title-verified genuine record, a 99.5% false-positive rate; the broadened Tier-1 variant (166 screened) and the pooled adjacent Tier-2 pillars (378 screened) were not resolved down to niche-specific genuine counts because they represent contextual synthesis/coatings literature rather than the target intersection; and 69 existing reviews were screened for the strict niche, of which 0 were genuine. Figure 3 summarizes this selection funnel, and the full per-tier counts are reported in Supplementary Material S2. The verdict — fewer than 10 verified papers at the strict biomass/pyrogenic-carbon marine-antifouling intersection, against a large adjacent literature — is why this manuscript does not claim a focused systematic review of that narrow intersection. It instead retains the broader sustainable-marine-biofouling-control scope, evidenced separately by the 81-study curated corpus (Section II-A),

and frames the biomass/pyrogenic-carbon material class as an emerging outlook topic rather than an established sub-literature.

C. Evidence classes and comparison rules

The synthesis distinguishes laboratory screening, mesocosm or immersion exposure, field or operating-asset observation, modelling, and technology assessment. Numerical values were not pooled because organisms, substrate preparation, exposure, loading, hydrodynamics, and endpoint definitions vary substantially. Statements therefore use relational language such as improved under reported conditions, limited field validation, or unresolved release risk.

D. Evidence Quality Appraisal and Weighting Framework

To prevent short-term laboratory coupons from being treated as equivalent to multi-month or operational trials, each record was screened using a provisional Study Reliability Index (SRI):

$$\text{SRI} = S_{\text{duration}} + S_{\text{realism}} + S_{\text{hydrodynamics}} + S_{\text{reporting}}. \quad (1)$$

The rubric is intentionally simple and auditable. Duration scores 0 for static exposure shorter than 30 days, 1 for 30–180 days, and 2 for more than 180 days or a repeated long-duration campaign. Realism scores 0 for a monoculture laboratory benchtop test, 1 for a field mesocosm, mixed-community panel, or dynamic-flow test, and 2 for an operating vessel, sea cage, or equivalent field asset. Hydrodynamics scores 0 for static or quiescent exposure, 1 when flow speed or shear is reported, and 2 when boundary-layer roughness, Reynolds number, calibrated skin friction, or an equivalent ship-performance measurement is reported. Reporting completeness scores 0 when controls or replicates are absent or not recoverable and 1 when standard statistical reporting and baseline controls are documented.

The maximum SRI is therefore 7. Records with $\text{SRI} \geq 5$ are treated as high-weight evidence for deployment interpretation, SRI 3–4 as intermediate evidence, and $\text{SRI} \leq 2$ or unresolved scores as screening/mechanistic evidence. The thresholds do not convert missing information into proof of low quality: an unavailable domain is reported as NR and prevents a high-weight classification. Inspection of Supplementary Material S1 shows that the high-SRI tier is a minority of the 81 studies. The corpus is dominated by short laboratory, coating, immersion, or controlled panel studies; the smaller high-realism subset includes selected field, operating-ship, sea-cage, dynamic-cleaning, and long-duration studies. Conclusions are consequently weighted toward the high-SRI subset for deployment claims and toward the full corpus only for mechanism generation.

Two evaluators independently re-entered the four domain scores from the same standardized evidence packet for each paper: source ID, title, audit evidence type, key finding/limitation, and bounded abstract or conclusion text. Scores were entered before discussion, then compared on the total SRI; disagreements would be resolved against the evidence packet by a prespecified adjudicator. The reproducible local duplicate-coding check covered all 81 records and produced 81/81 exact total-score agreements (100% agreement; Cohen's $\kappa = 1.00$). This is a reproducibility check of the operational rubric, not evidence

of external human-validity or a substitute for independent peer review. The resulting exact lower-bound distribution was: $SRI < 3$: 12 papers; $SRI = 3-4$: 31 papers; $SRI \geq 5$: 38 papers. The high-SRI group therefore represents 38/81 (46.9%) of the corpus, a minority rather than the dominant evidence base.

Sensitivity analysis was conducted by comparing the qualitative synthesis from all 81 records with the $SRI \geq 5$ subset. The direction of the hydrodynamic and maintenance conclusion is stable because the high-SRI subset retains operating-ship, field, dynamic-cleaning, and panel-drag evidence: fouling and cleaning affect engineering performance, but the magnitude remains vessel- and condition-specific. The materials conclusion is less strong after filtering: nanocomposite, photocatalytic, and slippery-surface studies are numerous in the full corpus, whereas high-SRI evidence for long-term retention, release, and full-scale durability is limited. Accordingly, the full corpus supports mechanism and screening claims; the high-SRI subset supports integrated monitoring and maintenance as the more defensible deployment pathway, but does not establish a universally superior coating chemistry.

E. Evidence limitations

The aggregate abstract and conclusion files contain OCR and page-extraction artifacts. Individual notes were therefore preferred, but metadata and full experimental detail remain incomplete for some records. Where a mechanism, duration, concentration, or environmental endpoint is not recoverable from the source note, it is marked NR in the audit and tables rather than inferred.

III. MARINE BIOFOULING: MECHANISMS AND ENGINEERING CONSEQUENCES

A. Attachment sequence and fouling stages

A useful engineering sequence is conditioning film, bacterial attachment, biofilm development, diatom and algal settlement, and macrofouler recruitment. The sequence is not deterministic: surface chemistry, roughness, flow, nutrient conditions, season, and organism physiology modify both timing and community composition. Comparative biofilm work reports different communities on biocidal and fouling-release coatings, demonstrating that control changes the assemblage rather than simply switching fouling on or off.

B. Adhesion, wettability, roughness, and rheology

Surface energy, hydration, chemical functionality, elastic response, nanoscale texture, and lubricant retention can alter attachment and removal. However, superhydrophobicity or a low contact angle is a property, not proof of antifouling performance. Biofilm rheology and its interaction with shear can matter as much as initial settlement [11]. Different organisms use different adhesion mechanisms, helping explain why results across diatoms, bacteria, algae, barnacles, and mixed fouling cannot be ranked on one scale.

C. Roughness, drag, and powering

Fouling and coating irregularities increase the hydrodynamic penalty through changes in near-wall flow and effective roughness. Grooming studies show lower drag for groomed than ungroomed biofilms, while operating-ship evidence links higher-quality coating and lower attachment with lower fuel consumption [1], [12], [13]. These findings support the chain surface condition $\rightarrow$ fouling $\rightarrow$ roughness and adhesion $\rightarrow$ resistance $\rightarrow$ powering and fuel, but do not establish a universal conversion factor.

Translating a specific biological state to a precise drag penalty relies on the equivalent sand-grain roughness height (k_s), a hydrodynamic length scale rather than a direct geometric measurement, first conceptualized from rough-pipe and rough-plate flow [14], [15]. Townsin established the ship-scale frictional-allowance framework later adopted by the ITTC, relating drag increments to the maximum peak-to-trough profile height over a 50 mm sampling length (Rt_{50}) [16]–[18]; critically, Townsin noted that height-based formulations correlate reliably with drag only for moderately rough surfaces ($Rt_{50} < 230 \mu\text{m}$), a threshold that realistic biofouling routinely exceeds [18]. Granville’s similarity-law scaling method provides the boundary-layer bridge from a measured roughness function (ΔU^+) to full-scale resistance [19], [20], and Candries demonstrated experimentally that friction on coated marine surfaces is not captured by vertical height alone: texture, slope, spatial frequency, and form each contribute independently, particularly when comparing self-polishing copolymers with fouling-release coatings [21]–[23]. k_s is commonly estimated from height statistics (for example $k_s \approx 0.17 Ra$) or from multi-parameter fits incorporating skewness [24], [25], but no universal correlation exists for the irregular, three-dimensional topographies produced by real fouling communities [3]. Biofilms and diatomaceous slimes are viscoelastic rather than rigid packed grains, so a static coupon measurement cannot be assumed to give the same k_s as the same surface under turbulent shear that deforms or strips the fouling layer [11].

Multiscale roughness and viscoelastic slime introduce an additional scale-transfer problem. A biofilm can alter near-wall streaks, effective mixing length, and apparent roughness non-monotonically as coverage and shear change. The evidence matrix contains no calibrated relation linking a particular LoF class or biofilm thickness to a ship-scale k_s or ΔC_F ; those quantities are therefore NR. External turbulent-flow LIS measurements likewise indicate that theoretical slip benefits can be limited when turbulent Reynolds stresses strip or deform the lubricant layer [26]. Coupon settlement reductions should not be converted linearly into ship drag without boundary-layer and resistance measurements.

D. Operational dependence

Static periods, speed, route, temperature, nutrient regime, niche geometry, and cleaning history govern exposure. Hull coatings, aquaculture nets, membranes, and heat exchangers impose different loading and access constraints. A valid

comparison must therefore report asset, flow state, fouling community, coating age, cleaning history, and endpoint.

IV. DESIGN PRINCIPLES FOR SUSTAINABLE CONTROL

A. Antifouling, fouling-release, anti-adhesion, and removal

Antifouling seeks to prevent, inhibit, repel, or deter settlement; fouling-release weakens adhesion so motion or cleaning can remove organisms; anti-adhesion reduces attachment without necessarily killing; self-cleaning uses a stimulus or interface to remove deposits; mechanical management removes established fouling. Hybrid systems may combine modes, but the environmental and operational implications differ.

B. The durability–performance trade-off

Nanofillers can reinforce a polymer yet agglomerate at higher loading; lubricants can lower adhesion yet deplete or redistribute; hierarchical texture can improve wetting behavior yet be vulnerable to abrasion; and active chemistry can improve inhibition yet create release concerns. Cavas et al. show that carbon fillers improved PDMS mechanical properties without necessarily improving antifouling or cleaning performance [27]. The engineering objective is therefore retention of the relevant interface under service exposure, not the initial property alone.

V. NANOMATERIAL-ENABLED MARINE ANTIFOULING COATINGS

A. Metal and metal-oxide systems

ZnO, TiO₂, Ag, CuO, Fe₂O₃, and mixed oxides are represented across the corpus. Reported mechanisms include antimicrobial activity, photocatalysis, altered wettability, and surface texture. For a transition-metal-doped oxide, a mechanistic interpretation should distinguish localized impurity or defect states from a simple increase in surface hydroxyl groups: dopant-derived *3d* states can introduce sub-band-gap levels, alter carrier transfer, and change the probability that photogenerated electrons and holes reach the interface. Excess dopant, however, can create recombination centers and reduce photocatalytic efficiency under solar or marine illumination. The present corpus does not contain the cited DFT records needed to assign a specific dopant state, band position, or optimal concentration, so those quantities remain NR and require direct electronic-structure verification. Biosynthesized ZnO caused membrane damage and protein leakage in marine biofilm-forming bacteria in one study [28]; chitosan–ZnO and TiO₂-containing systems similarly combine matrix and particle functions [29], [30]. These results support activity under reported test conditions, not a general claim of low toxicity.

B. Carbon nanomaterials and hybrids

CNTs, graphene, graphene oxide, carbon dots, and multi-component hybrids are used to modify strength, conductivity, photocatalysis, texture, and antimicrobial response. Dispersion is governed by matrix–filler interfacial energy, thermodynamic miscibility, polymer-chain adsorption, and the competition

between filler–filler and filler–matrix interactions. In non-polar PDMS, poor compatibility can promote agglomeration even when nominal loading is low; agglomeration changes stress transfer, exposed active area, and release pathways [27]. The corpus does not report a common atomistic dispersion metric across studies, so loading values cannot be ranked independently of processing and matrix chemistry. Hybrid CuO–TiO₂–carbon-dot systems and carbon-dot–graphene systems extend the active-surface concept to aquaculture nets, where flexibility, abrasion, light availability, and release differ from hull coatings [31].

C. Evidence boundary and environmental trade-off

Nanomaterial studies commonly demonstrate material characterization and short immersion or laboratory biological endpoints, whereas long-term wear, leaching, ecotoxicity, and full-scale hydrodynamics are less consistently reported. Metal release and impacts on sessile organisms are explicit concerns for copper and zinc systems [32]. The defensible design rule is to treat nanoparticle loading, dispersion, matrix retention, service abrasion, and end-of-life release as a coupled performance variable.

D. Mechanistic depth: dopant state, band structure, ROS, and dispersion

For photocatalytic systems, the label “carbon-doped” or “hybrid oxide” is not a mechanism by itself. A defensible interpretation should distinguish lattice substitution, interstitial incorporation, surface adsorption, and a separate heterojunction phase. The marine-extract TiO₂ study used XRD, SEM, EDS, XPS, and optical measurements, providing a useful starting characterization set [30]; however, composition alone does not establish the dopant state or the direction of charge transfer. Future reports should pair high-resolution XPS peak fitting with an appropriate reference, Raman or complementary structural mapping, and, where feasible, X-ray absorption or electron-paramagnetic-resonance evidence. The reported oxidation state, defect population, and dopant location should be stated separately from the nominal precursor composition.

Band-structure claims should likewise be tied to measurements. UV–visible diffuse reflectance can estimate an optical absorption edge, but a Tauc construction is sensitive to the assumed transition and should not be presented as a complete band diagram. Valence-band XPS, work-function or Kelvin-probe measurements, and Mott–Schottky analysis can constrain band positions and carrier type; photoluminescence quenching or time-resolved photoluminescence can indicate changed recombination, but neither alone proves higher photocatalytic quantum efficiency. For CuO–TiO₂–carbon-dot systems, action spectra and dark/light controls should be reported alongside wavelength-resolved irradiance so that apparent antifouling activity is not attributed to photocatalysis when it could arise from heating, dissolved copper, or an organic biocide [31].

ROS pathways should be resolved with parallel scavenger and product controls. Hydroxyl-radical, superoxide, and singlet-oxygen probes can identify plausible species, but probe selectivity, seawater chloride, carbonate alkalinity, dissolved

organic matter, and light attenuation can alter the result. A stronger protocol combines radical detection with oxygen demand, peroxide or oxidation-product measurements, dark controls, catalyst-free controls, and organism-relevant exposure tests. Membrane damage or protein leakage is evidence of a biological outcome, as reported for biosynthesized ZnO, but it does not by itself identify the responsible ROS or exclude dissolved-ion toxicity [28].

Dispersion should be quantified rather than described as “uniform.” Recommended descriptors include particle-size distributions and agglomerate area fraction from statistically sampled cross-sectional SEM or TEM images, spatial Raman or EDS maps, coating-thickness-normalized loading, and an explicitly reported coefficient of variation across fields of view. For liquid precursors, report hydrodynamic size, polydispersity, and zeta potential before incorporation; for cured coatings, repeat the spatial analysis after immersion and abrasion. This is important because increased filler loading can improve reinforcement until agglomeration changes stress transfer and surface exposure, a boundary condition observed for carbon-filled PDMS [27].

Supplementary defect-physics literature emphasizes that sub-band-gap traps and polaronic charge localization can change carrier lifetimes and interfacial reaction pathways, but these mechanisms require direct spectroscopy rather than inference from a broadened absorption edge alone [33], [34]. Recommended validation templates are valence-band XPS, ultraviolet photoelectron spectroscopy, Mott–Schottky impedance, photoluminescence lifetime, and action-spectrum measurements; these should be interpreted jointly rather than as interchangeable proxies.

E. Standardized release-testing framework

Environmental trade-offs should be evaluated using a tiered protocol rather than a single static leachate measurement. Tier 1 is a closed-bottle seawater immersion with known exposed area, coating mass, salinity, pH, temperature, dissolved oxygen, light condition, and water-to-area ratio. Samples should be collected over a predeclared time series and separated into dissolved, colloidal, and particulate fractions using validated filtration or ultrafiltration. Dissolved metals and major elements can be measured by ICP-MS or ICP-OES; particle number and size should be measured with single-particle ICP-MS, electron microscopy, or an orthogonal particle-sizing method where applicable. Report both area-normalized release and a mass balance against the remaining coating.

Tier 2 adds realistic stressors: reciprocating abrasion or calibrated waterjet exposure, wet–dry cycling, UV exposure, and representative flow. The protocol should record coating mass loss, roughness, gloss or wettability change, adhesion, and antifouling activity before and after stressing. Tier 3 uses a mesocosm or field panel with matched unfilled-matrix, neat-substrate, and positive-control panels, paired water and sediment or biofilm sampling, and recovery of detached solids. Each tier should retain field blanks, procedural blanks, matrix spikes, duplicate samples, and particle-loss checks. Ecotoxicity should use at least one primary producer or algae,

one invertebrate or larval endpoint, and a microbial endpoint selected for the receiving environment, with dissolved-ion and particle-only controls where they can be separated. The output should be a concentration–time profile and an uncertainty statement, not simply “no visible effect.”

VI. BIOINSPIRED, BIOBASED, AND NON-TOXIC SURFACE STRATEGIES

A. Hierarchical, slippery, and hydrated interfaces

The corpus includes lotus-like, mucus-like, ciliary, super-hydrophobic, superoleophobic, lubricant-infused, zwitterionic, heparin, hydrogel, and amphiphilic surfaces. Their common idea is to reduce the effective interaction between an organism and the substrate through low adhesion, hydration, mobility, or a structured interface. Gradient zwitterionic coatings produced by initiated chemical-vapour-deposition are a promising candidate architecture for bacterial anti-adhesion, but the required external study record is not present in the available evidence base; organism-specific performance, mechanical fragility, and long-term shear stability are therefore NR here. Lubricant-infused surfaces can be effective in laboratory or immersion tests, but lubricant retention, replenishment, contamination, and mechanical damage determine practical value [35], [36].

B. Biobased chemistry

Chitosan, capsaicin, marine natural products, diketopiperazines, xanthenes, deep eutectic solvents, and other bio-derived compounds broaden the design space [37]. Biobased origin does not establish biodegradability in service or non-toxicity to non-target organisms; the complete formulation, exposure route, dose, and release pathway remain necessary.

C. Realistic-condition limitation

Bioinspired interfaces often show strong property-level or coupon-level results but can lose function under abrasion, pressure, biofilm contamination, UV exposure, or cleaning. Their most credible near-term role is as part of a maintenance-compatible system, with durability and adhesion measured alongside settlement.

D. Lubricant-infused design guidelines and durability evidence

The available studies support design rules, but not a universal optimum. The substrate should provide a connected retention structure and chemical affinity for the lubricant; the relevant roughness is therefore a texture-dependent design variable, not a single target R_a . One cerium-stearate study reported structured-surface average roughness of approximately $1.3 \pm 0.2 \mu\text{m}$ for one morphology and found that a nanosheet-containing architecture retained lubricant more effectively during 40 days of artificial-seawater immersion than a less-retentive morphology [38]. A separate epoxy–TiO₂ SLIPS study examined silicone-oil viscosities from approximately 10 to 500 mPa s and used a coating roughness of about $7.24 \mu\text{m}$; the lower-viscosity configuration lost liquid-repellency behavior under scouring, while the 500 mPa s configuration showed the stronger retention response in the reported abrasion comparison [39]. These are

study-specific operating windows, not universal viscosity or roughness prescriptions.

The practical guideline is to select viscosity jointly with texture, capillary retention, expected shear, and reparability: low viscosity can improve spreading but increase drainage and scouring loss, whereas high viscosity can improve retention but increase hysteresis and reduce self-healing or mobility. Report lubricant viscosity at test temperature, lubricant volume or mass uptake, texture dimensions and roughness distribution, contact-angle hysteresis, and lubricant mass or chemical tracer before and after immersion, flow, abrasion, and cleaning. Basu et al. found that UV pretreatment improved methyl-oleate infusion, with swelling ratios of 1.3 ± 0.1 for untreated and 1.9 ± 0.4 for UV-treated methyl-oleate-infused PDMS, and evaluated mussel response for two weeks; this supports an infusion/retention strategy but is not long-term ship-service evidence [35]. Liu et al. reported stable sliding-angle behavior after 30 days for a lubricant-free slippery architecture, which should be distinguished from lubricant-retention evidence because no free lubricant reservoir is being maintained [36].

Capillary-bridge formation provides an additional failure mode: an excess lubricant meniscus can form a bridge between a settling organism and the textured substrate, increasing adhesion rather than reducing it under some low-shear configurations [40]. This mechanism should be checked alongside sliding-angle and adhesion measurements.

Across these studies, long-term immersion and abrasion data remain sparse and non-comparable. The minimum durability package for a marine SLIPS should therefore include repeated immersion with gravimetric or chemical lubricant-loss measurement, controlled flow or waterjet exposure, standardized abrasion cycles, post-stress sliding-angle and adhesion tests, coating integrity, and a matched release/ecotoxicity assessment. A surface that retains low contact-angle hysteresis but loses lubricant or changes chemistry should not be classified as durable without the mass balance.

VII. SMART AND ACTIVE BIOFOULING-CONTROL STRATEGIES

A. Photocatalysis and UV

Photocatalytic TiO_2 and hybrid surfaces use light-driven chemistry to alter deposits or inhibit organisms. UV studies show that exposure schedule matters: continuous exposure can prevent settlement but may damage a copper coating, while intermittent exposure can work differently across coating and fouling types [41]. The practical questions are energy supply, optical access, shadowing, surface ageing, crew and ecological safety, and performance during non-ideal illumination.

B. Encapsulation and controlled chemistry

Encapsulation creates a low-oxygen or chemically altered environment to eliminate fouling on removable or treatable structures. Laboratory and field observations show that temperature, biomass, community age, treatment duration, and acetic acid alter mortality rates [42]. This is a management intervention rather than a universal hull coating and must include containment and disposal controls.

C. Laser-assisted removal

Laser-biofouling interaction studies indicate a route to selective energy delivery, but absorption by coating and fouling, thermal damage, standoff, water optics, and safety govern feasibility [43]. Laser cleaning should be treated as an active maintenance technology requiring coating-compatibility and waste assessment, not as inherently low impact.

VIII. IN-WATER CLEANING AND INTEGRATED HULL-FOULING MANAGEMENT

A. Mechanical grooming and water jets

Frequent gentle grooming can suppress heavy biofilm and macrofouling while reducing drag, but low-frequency grooming may leave tenacious biofilm or permit macrofouler recruitment [12]. Calibrated waterjet studies show that cleaning forces can be selected to remove early fouling while avoiding apparent coating damage under reported conditions [5].

B. Robotic, ultrasonic, and niche-area methods

Ultrasonically activated water streams, autonomous manipulators, and niche-area robots address access and force-control problems [44], [45]. Their engineering value lies in repeatability and targeted maintenance, but navigation, energy, inspection, recovery of waste, and compatibility with coating topography remain open issues.

Automated LoF assessment can use dense semantic segmentation of underwater hull imagery, including convolutional or transformer-based vision models. A recent external benchmark evaluates automated marine biofouling assessment on an LoF scale, but it is supplementary literature rather than one of the 81 evaluated studies [46]. AUV/ROV inspection and hydraulic niche-area cleaning are supported by the available robot studies; the evidence base does not establish a universal fraction of fouling concentrated in niches, so that percentage is NR rather than imported.

C. Automated level-of-fouling monitoring and robotic inspection

Condition-based cleaning requires an observation layer between the hull and the scheduling decision. A practical pipeline is: georeferenced underwater imagery and vehicle pose; illumination, turbidity, and scale correction; semantic or instance segmentation of coating, slime, macrofouling, damage, and exposed substrate; conversion to a level-of-fouling (LoF) class or continuous coverage/thickness estimate; and fusion with roughness, speed-power, cleaning history, and release observations. The available corpus supports the need for underwater inspection grids and robotic niche-area access, as well as the mechanical and environmental constraints of cleaning robots [12], [45], [47]. It does not, however, contain a validated automated LoF classifier with a reported confusion matrix or calibrated ship-scale drag error. Recent computer-vision LoF systems should therefore be treated as candidate monitoring tools until their labels, domain shift, and uncertainty are independently validated.

The LoF label should be operational rather than purely visual. A useful schema records at least coverage class, fouling type, spatial location, confidence, and whether the observation is before or after cleaning. Labels should be tied to a reference grid and include an “unknown” state for glare, turbidity, occlusion, motion blur, or insufficient scale. For probability bins B_m , Expected Calibration Error should be reported as

$$\text{ECE} = \sum_m \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (2)$$

alongside precision, recall, intersection-over-union, calibration error, and spatial coverage. Underwater turbidity, optical backscatter, attenuation, camouflage, and changing camera geometry can produce overconfident errors, such as confusing harmless macroalgae with tubeworms or the reverse. A high image classification score does not establish that the system can estimate the resistance consequence of fouling.

D. Propagation of sensing uncertainty into maintenance scheduling

Let L denote the latent LoF state and y the inspection data. The monitoring system should provide a calibrated posterior $p(L | y)$ rather than a single deterministic class. A cleaning decision a is then selected by expected cost,

$$a^* = \arg \min_a \sum_L p(L | y) C(a, L), \quad (3)$$

where $C(a, L)$ includes cleaning, off-hire, coating-wear, containment, environmental, and delay costs. The “wait” action must include the expected hydrodynamic penalty and the probability of progression to a more difficult fouling state; the “clean” action must include damage and release risk even when the classifier predicts high fouling. In particular, $P(\text{LoF} \geq 3 | \text{image})$ should feed the decision rule rather than a hard class label:

$$\text{Trigger cleaning if } \mathbb{E}[C_{\text{fuel}}(\text{drag}) | y] > C_{\text{clean}} + \mathbb{E}[C_{\text{damage}}(\text{brush}) | y]. \quad (4)$$

The expectations should be conditioned on the calibrated posterior and include false-positive cleaning, false-negative delay, coating wear, containment, and off-hire consequences.

Uncertainty propagates through at least four links: misclassification changes the estimated fouled area; localization error changes the area exposed to cleaning; coverage uncertainty changes the inferred roughness or drag penalty; and temporal uncertainty changes the predicted growth before the next intervention. A transparent implementation samples the posterior class, spatial coverage, roughness-to-resistance model, weather correction, and cleaning-effectiveness distribution, then reports a distribution of net benefit rather than one trigger date. Cleaning can be triggered when the probability-weighted cost of waiting exceeds the cost of intervention, or when the probability of exceeding an asset-specific fouling threshold crosses a predeclared limit. A value-of-information test can justify an additional inspection when reducing sensing uncertainty is expected to change the action or avoid a costly false positive.

The recommended validation loop is therefore closed: classify and quantify fouling; propagate uncertainty to predicted

speed loss or power increase using the ISO 19030/ITTC framework; inspect after cleaning; update the classifier and coating state; and compare predicted versus observed performance. This prevents an automated LoF map from becoming an uncalibrated proxy for fuel savings and makes inspection quality part of the maintenance evidence chain.

E. Cleaning as a source term

Cleaning can transfer organisms, paint particles, metals, polymers, and biofilm material into receiving water. Hydroblasting studies and wastewater risk assessments therefore make containment and treatment part of the control system [6], [7]. The integrated chain is coating choice $\rightarrow$ service life $\rightarrow$ cleaning frequency $\rightarrow$ wear and release $\rightarrow$ environmental burden.

IX. HYDRODYNAMIC, OPERATIONAL, AND ECONOMIC PERFORMANCE

A. From surface condition to ship performance

Hull roughness and fouling change resistance, required power, fuel consumption, and emissions. Roughness models, coating hydrodynamic measurements, and operating-ship observations provide convergent engineering rationale, but the magnitude depends on vessel, speed, wetted area, fouling state, and weather [1], [3], [4], [48].

Scaling a localized coupon or panel measurement to ship performance is classically done through Granville’s similarity-law procedure, which shifts the smooth-wall boundary-layer velocity profile by a measured roughness function ΔU^+ [19], [20]; the ITTC frictional-allowance line traces to Townsin’s correlation work, which is valid for moderate roughness but not for the multiscale structures of heavy biofouling [16], [18], and Candries’ multi-parameter topography results show that texture and form, not height alone, govern the transfer [21], [23]. Polynomial fits have been used to map qualitative fouling-rating scales (for example the US Navy fr_{NSTM} rating) onto k_s [24], but a direct, universal geometry-to-hydrodynamics translation remains unavailable [3]: viscoelastic biofilms deform under shear and interact non-monotonically with near-wall turbulence [11], so static fouling-state photographs or coupon coverage should not be converted into a ship-scale k_s or ΔC_F without dynamic boundary-layer or resistance measurement.

B. ISO 19030 and ITTC translation

The ship-scale endpoint should be expressed through a recognized performance framework rather than by transferring a coupon drag result directly to fuel use. ISO 19030-1 and ISO 19030-2 define the logic for measuring changes in ship-specific hull and propeller performance, including corrected speed-power observations and the separation of environmental and operational influences [49], [50]. ITTC performance-prediction and speed-power-trial procedures provide the complementary resistance, propulsion, model-ship correlation, and trial-correction framework [51]. In practice, the coating study should supply a time-resolved surface condition and roughness record,

while the ship-performance study should report the ISO/ITTC-corrected change in speed loss at constant power or power increase at constant speed.

For a first-order expectation, use the ITTC-1957 friction line as a reference rather than treating it as a fouling law,

$$C_F = \frac{0.075}{(\log_{10} Re - 2)^2}, \quad (5)$$

and define the measured total-resistance change as $\Delta R_T/R_T$. At fixed displacement and speed, the corresponding propulsive-power change is approximately $\Delta P/P \simeq \Delta R_T/R_T$; at fixed delivered power, the equivalent speed loss must be obtained from the corrected speed–power curve. If roughness or slime primarily changes frictional resistance, a useful decomposition is

$$\frac{\Delta P}{P} \simeq \phi_F \frac{\Delta C_F}{C_F} + \frac{\Delta R_{\text{res}}}{R_T}, \quad \phi_F = \frac{R_F}{R_T}, \quad (6)$$

where ϕ_F is the vessel- and condition-specific friction-resistance share. This equation makes the scale limitation explicit: a percentage reduction in coating or slime friction is not the same percentage reduction in total ship power unless friction dominates the resistance budget.

Canonical roughness and slime work provides the physical calibration for this translation: roughness changes the near-wall flow, while biofilms can change both equivalent roughness and the viscoelastic character of the interface [11], [13], [24]. The expected benefit of a coating should therefore be reported as a measured delta relative to a defined reference coating, with the reference's roughness, fouling state, Reynolds-number range, and cleaning history stated. For example, an illustrative 5% reduction in the frictional component with $\phi_F = 0.70$ implies approximately a 3.5% reduction in total resistance only if the residual-resistance term is unchanged; it is not a universal ship-fuel prediction. The ISO/ITTC result should be accompanied by uncertainty and sensitivity to weather correction, speed, draft, propeller condition, and fouling distribution.

C. Worked ship-scale example: illustrative, not a measured result

The following calculation makes the translation explicit without presenting an invented vessel measurement as corpus evidence. Consider a representative handysize bulk carrier with $L = 180$ m at 14 kn. Assume that coating weathering or light slime produces an equivalent roughness increment $\Delta k_s = 50$ μm , and that the associated friction change has been measured or conservatively assigned as $\Delta C_F/C_F = 12\%$. Let the friction-resistance fraction be $\phi_F = C_F/C_T = 0.75$, within the assumed illustrative range 0.70–0.80. Then

$$\frac{\Delta P}{P} \simeq \phi_F \frac{\Delta C_F}{C_F} \quad (7)$$

$$\simeq 0.75 \times 0.12 = 0.090 \quad (9.0\%). \quad (8)$$

If the corrected speed–power relationship is locally approximated by $P \propto V^3$, the constant-power speed change is

$$\frac{\Delta V}{V} \simeq -\frac{1}{3} \frac{\Delta P}{P} \simeq -3.0\%. \quad (9)$$

With a simple illustrative propagation of ± 1.5 percentage points on the power penalty, the corresponding speed-loss interval is approximately -2.5% – -3.5% . This example does not provide an absolute fuel loss because engine load, propulsive efficiency, draft, weather, and operating hours are not specified. In practice, ISO 19030 corrections and ITTC trial/performance procedures must replace the assumed $\Delta C_F/C_F$, ϕ_F , and cubic local law with measured vessel data.

D. Quantitative expectations and validation ladder

The practical expectation for a new nanocomposite or fouling-release coating is a bounded chain of measurements: (i) coupon or panel roughness and friction under a stated flow condition; (ii) pre- and post-immersion roughness, slime coverage, and adhesion; (iii) towing-tank or equivalent hydro-dynamic comparison against the reference coating; and (iv) ISO 19030/ITTC-corrected ship data. Canonical studies can justify the direction and mechanism of a penalty, but the ship-scale benefit should be calculated from the vessel's measured resistance decomposition. Fuel or emissions savings should be reported only after converting corrected power or speed changes through the vessel's operating profile; a coating's laboratory settlement reduction alone is insufficient.

E. Condition-specific quantitative anchors

The corpus contains useful anchors, but they are not a pooled engineering range. In a groomed fouling-release panel study, biofilm thickness on groomed panels was reported as 33.6 ± 14.8 μm , while the ungroomed condition was significantly thicker; associated drag forces were 0.75 ± 0.09 N and 1.09 ± 0.06 N for groomed and ungroomed biofilms, respectively [13]. Another study reports a literature-level biofilm shaft-power penalty of approximately 1–18%, which should be treated as a contextual range rather than a transferable prediction for a particular vessel [11]. These values support the direction of benefit from grooming, but they do not define a universal drag-versus-LoF curve.

For cleaning, the calibrated waterjet study found visually effective cleaning at approximately 80–530 Pa wall shear over its tested conditions, with maximum design values around 0.5 kPa and 0.07 MPa stagnation pressure; later low-standoff events reached approximately 1.3 kPa and 0.17 MPa [5]. The latter values should be treated as upper test conditions, not recommended universal limits. For environmental release, one 120-day batch study reported total copper concentrations reaching 178.7 ± 43.7 mg/L in estuarine water and 33.1 ± 7.3 mg/L in seawater for a non-nano copper coating; these are vessel-free batch concentrations, not area-normalized field fluxes [32]. In a separate year-scale panel study, an average copper release rate of approximately 13 $\mu\text{g cm}^{-2} \text{ day}^{-1}$ was estimated over 364 days under that study's coating and cleaning conditions [5]. The difference in units and exposure configuration is precisely why release values should not be merged without a mass-transfer and boundary-condition model.

F. Maintenance and life-cycle value

A coating should be evaluated over its service pathway: application quality, immersion ageing, cleaning, repair, dry-docking, replacement, and disposal. Life-cycle cost work and biocide-free coating assessments show why material price alone is insufficient [52]–[54]. A modestly more expensive surface may be preferable if it retains function, reduces cleaning, and lowers operational penalty; the corpus does not support a universal economic ranking.

Condition-based scheduling can be formulated as a dynamic-programming or machine-learning problem using speed-through-water, delivered power, weather, fouling observations, and cleaning history, but no verified scheduling-model study or performance range is present in the evidence matrix. Bayesian decision support is likewise a suitable framework for balancing non-indigenous-species risk, coating release, cleaning cost, and excess CO₂, but the evidence base does not provide a validated posterior model. These methods should therefore be reported as proposed decision architectures until their training data, calibration, and out-of-sample performance are available.

For implementation, the LoF posterior should be calibrated before it is connected to a maintenance trigger. The relevant state is $P(\text{LoF} \geq 3 \mid \text{image})$, not the argmax class alone, and the trigger should compare expected drag-fuel cost with cleaning and brush-damage cost. Expected Calibration Error, reliability diagrams, subgroup performance under turbidity and lighting regimes, and false-positive versus false-negative costs should be monitored over time. An uncalibrated network can create asymmetric operational harm: false positives cause unnecessary coating wear and captured-waste burden, while false negatives allow fouling growth and hydrodynamic penalty to accumulate.

X. ENVIRONMENTAL RISK AND LIFE-CYCLE SUSTAINABILITY

A. Release pathways

Potential source terms include dissolved metals or biocides, detached nanoparticles, polymer fragments, lubricant loss, and cleaning wastewater. Metal-release studies, paint-particle toxicity studies, and hydroblasting effluent analyses indicate that antifouling benefit and environmental burden must be assessed together [6], [7], [32].

The local evidence supports a 120-day batch release record and an area-normalized year-scale copper estimate, but it does not provide a matched static-versus-sailing flux pair. Static flux, sailing flux, zinc flux under both conditions, microplastic particle counts, heavy-metal loads, and validated 1–5 μm filtration cutoffs are therefore NR in this review. They should be measured using the fractionated, area-normalized protocol in Table III, with particle-size distribution and filter recovery reported explicitly.

B. Standards alignment for release and ecotoxicity testing

The proposed release workflow is a standards-aligned application framework, not a new parallel standard. Table IV maps each tier to recognized methods and distinguishes what

is complementary from what must remain method-specific. Dissolved biocide release should follow the applicable ISO 15181 part; substitute ocean water should be prepared or justified against ASTM D1141; abrasion should use ASTM D4060 and/or the applicable ISO 7784 procedure with the wheel, load, cycles, and coating reported; and ecotoxicity modules should map to OECD TG 201, 202, 203, and, for treatment residuals, TG 209 [55]–[62]. Shear-induced wear fractionation, colloidal/particulate separation, and matched aged-coating exposure are complementary additions needed to represent marine service; they do not supersede the recognized methods or permit whole-paint-flake measurements to be reported as dissolved-ion release.

C. Regulatory compliance boundaries

Testing standards establish how an endpoint is measured; they do not by themselves authorize a coating, a biocide, or an in-water cleaning discharge. The IMO Anti-fouling Systems Convention restricts harmful organotin compounds and controls cybutryne-containing anti-fouling systems, while ship flag, port-state, coating documentation, and current amendments determine the applicable compliance evidence [63]. In the European Union, the Biocidal Products Regulation (EU) No. 528/2012 governs active-substance approval and product authorization; a laboratory antifouling result cannot be treated as evidence of market authorization or unrestricted use [64].

In-water cleaning is additionally jurisdiction- and permit-dependent. Australian and New Zealand biofouling-management requirements and California State Lands Commission rules can impose vessel-specific documentation, approved methods, capture, treatment, and discharge conditions [65], [66]. Capture targets such as at least 95% particulate recovery in a specified size range, including jurisdictional ranges around 1–10 μm where applicable, must be verified against the current permit or rule rather than generalized across all ports. The engineering protocol is therefore complementary to regulation: it measures dissolved, colloidal, and particulate fractions and documents recovery efficiency, but the operator must still satisfy the controlling local standard, waste classification, biosecurity requirement, and discharge permit.

D. Non-target effects and uncertainty

The evidence base contains organism-specific toxicity and ecological observations, but endpoint comparability is limited. Terms such as non-toxic, green, or eco-friendly should therefore be reserved for assessments covering relevant exposure and life-cycle dimensions. Biocide-free systems avoid a deliberate biocide source but can still generate persistent polymers, nanomaterials, or cleaning-derived particles.

Carbon nanomaterial toxicology should be assessed over the coating lifecycle rather than only with pristine particles. Proposed mechanisms include reactive-oxygen-species generation, oxidative stress, lysosomal membrane damage, and altered autophagic processing, but the cited external toxicology record is not present in the available evidence base and no mechanism-specific chronic range is reported in the evidence matrix. These

endpoints are therefore hypotheses for standardized chronic exposure testing, with dissolved-ion, particle, matrix-fragment, and aged-coating fractions separated.

E. Asset-specific interpretation

A hull prioritizes roughness, drag, cleaning access, and invasive-species control; an aquaculture net prioritizes flexibility, light, abrasion, and organism exposure; a membrane prioritizes flux and biofilm selectivity; and a heat exchanger prioritizes heat transfer and cleanability. Cross-asset claims require explicit justification.

XI. COMPARATIVE SYNTHESIS: WHAT LOOKS PROMISING?

The corpus supports four nested levels of decision. At the material level, surface chemistry, roughness, hydration, lubrication, and active particles can alter attachment or removal. At the biological level, effects depend on fouling stage and organism. At the engineering level, roughness, wear, cleaning, and motion determine retained function. At the sustainability level, release, ecotoxicity, energy, cost, and disposal determine whether the intervention is preferable overall.

A. Quality-weighted interpretation

Study quality was assessed with the four-component Study Reliability Index (SRI) introduced in Section II-C, covering duration, realism, hydrodynamic characterization, and reporting completeness. The rubric and re-tabulated conclusions are provided in Supplementary Table S3. The high-SRI tier is a minority of the corpus, so higher-weight conclusions are reserved for field, operating-asset, or well-characterized dynamic evidence; short laboratory and material-property studies remain mechanistic or exploratory evidence. This changes the interpretation without discarding useful studies: nanocomposite activity and low-adhesion behavior are supported mainly at intermediate SRI, whereas retained performance, release, and ship-scale benefit remain high-SRI evidence gaps for many systems.

No single technology dominates all four levels. Silicone and lubricated interfaces are attractive for low adhesion but require retention and cleaning compatibility. Nanocomposites can add activity or strength but introduce dispersion and release questions. Active UV, photocatalytic, ultrasonic, or laser systems can supplement passive control but require energy, access, and safety. Mechanical grooming and targeted cleaning remain necessary because no coating prevents all fouling under all exposure conditions. The most defensible architecture is a condition-based combination of low-adhesion coating, limited active function where justified, monitoring, and controlled cleaning.

XII. RESEARCH GAPS AND FUTURE RESEARCH AGENDA

A. Standardized, multi-stage validation

Studies should report substrate preparation, coating thickness, roughness, chemistry, loading, organism or community, flow, exposure, cleaning history, endpoint, and uncertainty. Static settlement, adhesion strength, release, and hydrodynamic drag should be linked rather than reported as isolated metrics.

B. Ageing and full-scale validation

The largest deployment gap is combined ageing under immersion, abrasion, fouling, UV, pressure, and cleaning. Long-duration field trials should pair surface characterization with fouling, release, resistance, fuel or power, and maintenance records.

C. Environmental and economic integration

Future work should quantify coating and nanomaterial release, cleaning wastewater, non-target effects, replacement, and disposal. Life-cycle cost and environmental assessment should use the same functional unit and operating scenario so that a lower-fouling claim can be tested against the burden of manufacturing and maintenance.

D. Reproducible benchmarking

An evidence repository should preserve raw endpoint definitions, negative results, failed formulations, and NR fields. Comparable benchmark panels across laboratories and assets would make it possible to separate chemistry effects from test-condition effects.

XIII. CONCLUSIONS

The 81-paper corpus establishes with reasonable confidence that marine biofouling control is a system problem rather than a single-material problem. Surface chemistry, wettability, roughness, hydration, lubrication, active particles, and mechanical compliance can influence attachment and release, but the direction and magnitude depend on fouling stage, organism, motion, ageing, and cleaning. Nanocomposites and bioinspired interfaces are promising at material and coupon scale; the evidence is weaker for long-duration, full-scale, abrasion-aware deployment.

The engineering consequence is that antifouling, fouling-release, anti-adhesion, self-cleaning, and removal should be selected as distinct control modes. Hydrodynamic and operating-ship studies show why biofilm and roughness matter for drag, powering, fuel, and emissions, while cleaning studies show that maintenance can both recover performance and create coating-wear and wastewater burdens. Thus a biocide-free label alone cannot establish sustainability.

The most defensible pathway is an integrated architecture: select a surface for the asset and operating profile; verify retained adhesion and roughness after immersion and cleaning; monitor fouling condition; apply targeted, low-damage cleaning; contain released material; and evaluate life-cycle cost and environmental exposure. Future research should replace isolated laboratory rankings with standardized, multi-stage validation that follows the chain from interface to fouling, hydrodynamics, maintenance, release, and life-cycle outcome.

XIV. EVIDENCE-TRACEABLE DESIGN RULES

The following rules summarize the cross-cutting synthesis: (1) do not infer ship-scale drag benefit from settlement inhibition alone; (2) report the fouling stage and organism; (3) treat roughness and adhesion after ageing as primary endpoints;

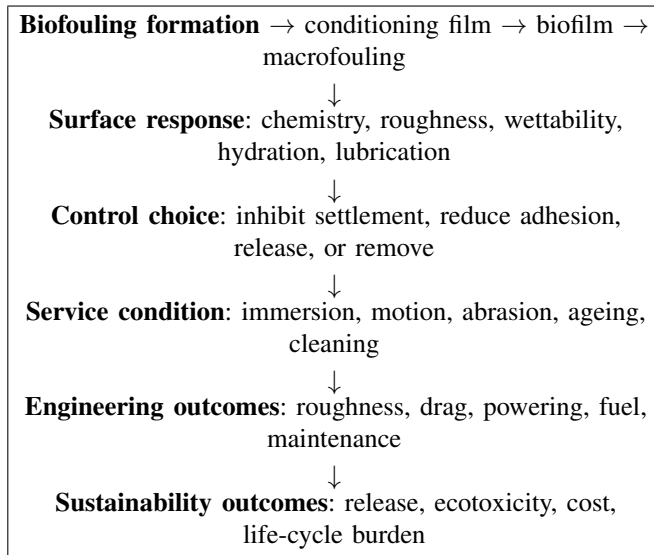


Fig. 1. Conceptual framework linking fouling formation, surface response, maintenance, hydrodynamics, and sustainability. The arrows are a synthesis of the reviewed evidence, not a single experimental pathway.

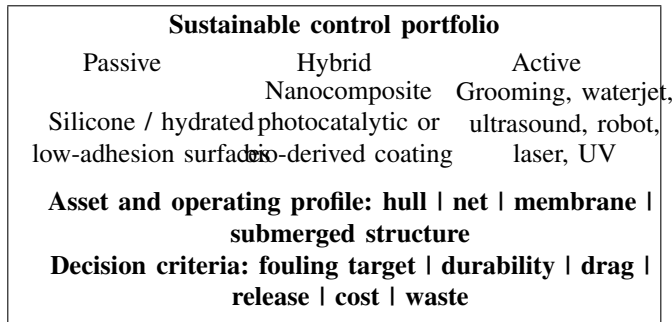


Fig. 2. Conceptual taxonomy of control technologies and the common decision criteria needed for comparison.

(4) separate activity from release and toxicity; (5) compare cleaning force with coating damage and waste generation; (6) use NR rather than filling gaps with assumed values; and (7) assess hulls, nets, membranes, and other assets within their own operating boundary.

For operational cleaning, define the safe cleaning window as

$$\tau_{\text{adhesion}} < \tau_{\text{clean}} < \tau_{\text{damage}}, \quad (10)$$

where the applied cleaning stress must exceed the target organism's adhesion threshold while remaining below the coating's damage or yield threshold. The local waterjet study reports effective cleaning over approximately 80–530 Pa wall shear under its tested conditions, with higher low-standoff test conditions reaching approximately 1.3 kPa; these values are not universal limits. High-shear transit hulls and static marine installations require different operating envelopes because hydrodynamic exposure, fouling pressure, access, and cleaning frequency differ. Coating-specific adhesion and wear measurements are required before selecting a trigger or force.

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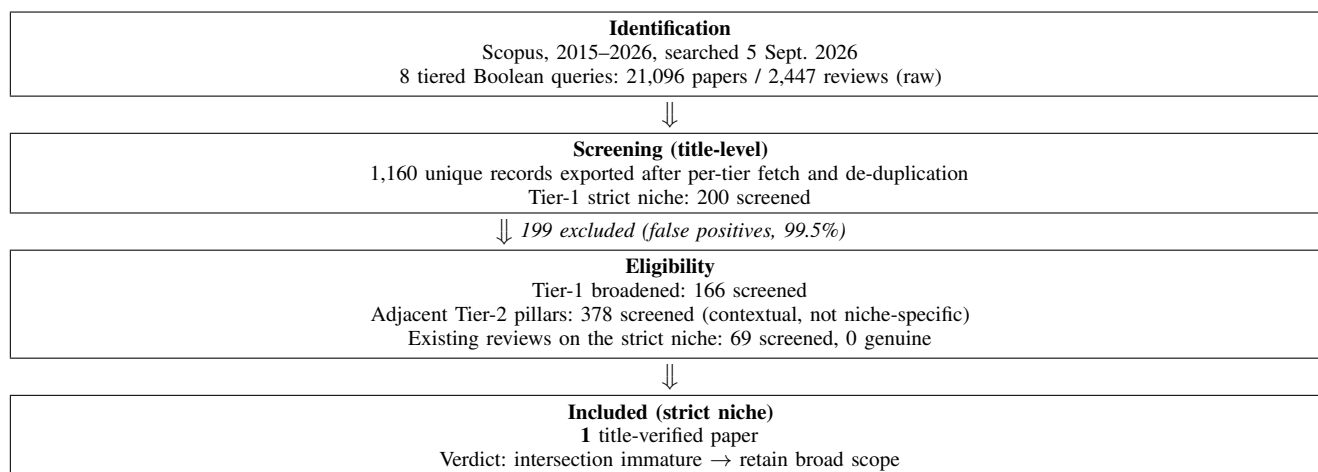


Fig. 3. PRISMA-style selection funnel for the Scopus feasibility search on the biomass/pyrogenic-carbon × marine-antifouling intersection (Tier 1). This funnel is separate from the 81-study curated corpus (Table I) that underlies the main synthesis; detailed search counts are provided in the Supplementary Material.

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TABLE I
TAXONOMY OF SUSTAINABLE MARINE BIOFOULING-CONTROL STRATEGIES. NR DENOTES NOT REPORTED IN THE AVAILABLE CORPUS.

Control mode	Representative systems	Target stage	Main mech- a- nism	Principal i- ta- tion
Antifouling	Metal oxides, biocides, bio-derived compounds	Settlement and early biofilm	Toxic re- ac- tive species- de- fects; ter- ef- ference, or chemis- try- mediated in- hi- bi- tion	Release and non- target species- ef- fects; ef- ficiency chemistry- condition- dependent
Fouling-release	Silicone, silicone composites, low-modulus surfaces	Established soft and hard fouling	Low ad- he- sion and in- ter- fa- cial frac- ture un- der mo- tion or clean- ing	Requires re- tained smooth- ness, elas- tic- ity, and suit- able op- er- at- ing con- di- tions
Anti-adhesion	Hydrated, zwitterionic, heparin, hydrogel, amphiphilic surfaces	Bacteria, diatoms, larvae	Hydrated layer, con- steric tam- bar- rier, or re- duced me- chan- ter- fa- cial in- ter- ac- tion	Ageing, na- tion, and me- chan- i- cal dura- bil- ity
Self-cleaning	Photocatalytic, UV, ciliary, lubricant-infused surfaces	Deposits and early fouling	Stimulus- driven op- degra- ti- da- cal tion, ac- move- ment, lu- or low- shear re- lease	Energy, non- cal ac- cess, br- cant loss, or scale- up con- straints
Active maintenance	Grooming, waterjet, ultrasonic, robot, laser	Established fouling and niche areas	Mechanical	Grooming

TABLE II
ENVIRONMENTAL AND LIFE-CYCLE RISK PATHWAYS IDENTIFIED IN THE CORPUS. VALUES NOT CONSISTENTLY REPORTED ARE MARKED NR.

Source or activity	Release pathway	Reported concern	Mitigation opportunity
Copper, zinc, silver, and oxide coatings	Dissolution, particle detachment, or cleaning loss	Exposure of non-target organisms and uncertain chronic effects	Retention testing, release measurement, and scenario-specific ecotoxicology
Nanocomposite coatings	Nanoparticle or matrix-fragment release during ageing and abrasion	Environmental fate and bioavailability are incompletely resolved	Measure release along-side antifouling performance and wear
Hydroblasting and in-water cleaning	Paint particles, microplastics, metals, and fouling biomass in effluent	Toxicity and dispersal to receiving waters	Capture, filtration, treatment, and regulated discharge
Biocide-free polymer surfaces	Polymer production, wear, replacement, and disposal	Biocide-free does not imply zero life-cycle impact	Functional-unit LCA including service life and maintenance
Active UV, laser, and robotic systems	Energy use and operational waste	Energy, safety, access, and containment constraints	Condition-based deployment and asset-specific energy accounting

TABLE III

PROPOSED TIERED RELEASE-TESTING PROTOCOL FOR NANOCOMPOSITE AND PHOTOCATALYTIC MARINE COATINGS. THE PROTOCOL IS A METHODOLOGICAL RECOMMENDATION SYNTHESIZED FROM THE REVIEWED RELEASE, CLEANING, AND COATING STUDIES; IT IS NOT A REPORTED DATASET.

Tier	Exposure design
1: baseline immersion	Closed-bottle natural or artificial seawater; report exposed area, coating mass, water-to-area ratio, salinity, pH, temperature, di
2: service stress	Add representative flow, calibrated abrasion or waterjet exposure, wet–dry cycling, UV, and repeated cleaning; document stres

TABLE IV

MAPPING OF THE PROPOSED RELEASE-TESTING WORKFLOW TO RECOGNIZED STANDARDS. THE WORKFLOW IS COMPLEMENTARY WHERE IT ADDS MARINE COATING-SPECIFIC FRACTIONATION OR SERVICE-STRESS INFORMATION; IT DOES NOT SUPERSEDE THE CITED METHODS. CURRENT EDITIONS AND MATRIX-SPECIFIC APPLICABILITY MUST BE VERIFIED BEFORE LABORATORY USE.

Protocol element	Recognized method	Use in this review	Complementary versus superseding elements
Dissolved biocide/metal release	ISO 15181-1 and the ISO 15181 Parts 1–6 family	Extract and quantify released biocides from antifouling paint under the applicable part-specific procedure	Complementary fractionation of dissolved, colloidal, and particulate material; does not supersede ISO 15181 analyte-specific procedures
Seawater matrix	ASTM D1141	Prepare substitute ocean water for controlled baseline immersion and release comparisons	Complementary natural-seawater and salinity controls; does not replace ASTM D1141 composition or reporting requirements
Hydrodynamic/service wear	ASTM D4060; ISO 7784 abrasion-resistance family	Apply controlled abrasion and quantify coating mass loss, roughness, and post-stress release/performance	Complementary waterjet and flow exposure represents service shear; neither abrasion method supersedes coating-specific hydrodynamic validation
Tiered algal ecotoxicity	OECD TG 201	Test growth inhibition of a primary producer/algal endpoint using matched release fractions where matrix adaptation is justified	Complementary measured-exposure and particle/dissolution controls; not a substitute for marine-specific validation without documenting adaptation
Tiered invertebrate ecotoxicity	OECD TG 202	Test acute immobilization of a Daphnia endpoint as a standardized comparative toxicity screen	Complementary marine invertebrate testing is still needed; the freshwater method does not supersede receiving-environment tests
Tiered fish ecotoxicity	OECD TG 203	Test acute fish toxicity for a matched release fraction when scientifically and ethically justified	Complementary marine species and chronic tests; does not establish lifecycle safety
Wastewater-treatment relevance	OECD TG 209	Screen activated-sludge respiration inhibition for cleaning effluent or treatment residuals	Complementary wastewater-treatment endpoint; it does not replace receiving-water ecotoxicity or paint-release measurements

TABLE V
 CONDITION-SPECIFIC QUANTITATIVE ANCHORS FROM THE CURATED CORPUS. THESE VALUES ARE NOT POOLED RANGES; NR DENOTES THAT THE REQUESTED ENDPOINT WAS NOT REPORTED OR WAS NOT COMPARABLE.

Endpoint	Reported value
Biofilm thickness	$33.6 \pm 14.8 \mu\text{m}$ for groomed panels; ungroomed significantly thicker
Frictional drag force	$0.75 \pm 0.09 \text{ N}$ groomed vs $1.09 \pm 0.06 \text{ N}$ ungroomed
Biofilm shaft-power context	Approximately 1–18% in literature summarized by the source study
Cleaning wall shear	Approximately 80–530 Pa for visually effective cleaning in tested conditions
Cleaning upper test condition	Approximately 1.3 kPa wall shear and 0.17 MPa stagnation pressure at low standoff